Machine-checked degree lower bounds for interpolating the elliptic curve discrete logarithm from x-coordinates

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Abstract

We record three degree lower bounds for polynomials that interpolate functions of the discrete logarithm from the x-coordinates of multiples kP of a point on a short Weierstrass curve, and we give Lean 4 formalizations of them. (B) If $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, where P has order r, then $r \le 6 \deg g + 5$. (C) If $g(x(kP)) = k$ for $1 \le k \le N$ and $2N < r$, then $N \le 3 \deg g + 2$. (D) If the interpolation $g(x(kP)) = k^2$ is required only on a set $A \subseteq [1, N]$, $2N < r$, then for each $j \ge 1$ at most $3 \deg g$ elements $k \in A$ have $k - j, k + j \in A$ (under an explicit extra hypothesis). All three are proved by one device, a cleared second-difference polynomial of exact degree $3 \deg g$. The statements are verified by Lean against a pinned Mathlib; the Lean statements are quoted verbatim. Closely related degree bounds are due to Lange–Winterhof and Satoh; we could not access their exact statements, so our bounds may be known or weaker. The contribution claimed is the machine-checked formalization and the precise hypotheses. The results concern polynomial degree in a restricted model and have no bit-complexity consequence.

1 Introduction

The elliptic curve discrete logarithm problem (ECDLP) asks, given P and $Q = kP$ in a group of order r, for k modulo r. A natural question about restricted classes of algorithms is whether the map $x(kP) \mapsto k$ (or a quantity such as k^2, which is the natural x-only invariant because $x(kP) = x(-kP)$) can be given by a low-degree polynomial. A lower bound on the degree of any interpolating polynomial shows that no *low-degree-polynomial formula* computes the logarithm; it is a barrier for one restricted model, in the spirit of the interpolation results cited in Section 6.

What such a bound does *not* say should be stated plainly. It says nothing about algorithms that use other operations (comparisons, branching, group operations, index calculus), nothing about the existence of an interpolating polynomial, and nothing about bit complexity: a polynomial of degree d over $\mathbb{F}_p$ can be evaluated with a number of field operations that is not controlled by d in general, and we make no statement relating d to running time. The open target of the laboratory (a universal classical ECDLP algorithm with worst-case $O(n \log n)$ bit operations) is unaffected by this report.

Contents. Section 2 fixes notation; Section 3 states the three results; Section 4 gives the proofs; Section 5 quotes the Lean statements and the trust base; Section 6 discusses prior work and novelty; Section 7 lists limitations.

2 Setting and notation

Let F be a field and $W : y^2 = x^3 + ax + b$ a short Weierstrass cubic over F ($a_1 = a_2 = a_3 = 0$, $a = a_4$, $b = a_6$). The principal case is $F = \mathbb{F}_p$ and W an elliptic curve with a point P of prime

order r. For a nonzero affine point Q we write $x(Q)$ for its x-coordinate, and we write $r = \mathrm{ord}(P)$ for the additive order of P. For a polynomial $g \in F[X]$ we write $m = \deg g$ and $\mathrm{lc}(g)$ for its leading coefficient. Throughout we assume $2 \neq 0$ in F.

Two facts are used. First, if $2N < r$ then $k \mapsto x(kP)$ is injective on $\{1, \dots, N\}$: $x(kP) = x(lP)$ forces $kP = \pm lP$, i.e. $k \equiv \pm l \pmod r$, and $2 \le k + l \le 2N < r$ rules out $k \equiv -l$, so $k \equiv l$, hence $k = l$. Second, the x-addition identities: for $Q = (u, v)$, $P = (x_0, y_0)$ affine points with $u \neq x_0$,

$$\big(x(Q+P) + x(Q-P)\big)(u - x_0)^2 = 2\big((x_0 u + a)(u + x_0) + 2b\big), \tag{1}$$

$$x(Q+P)\, x(Q-P)\, (u - x_0)^2 = (x_0 u - a)^2 - 4b(u + x_0). \tag{2}$$

For (1) write $s_\pm = x(Q \pm P) = \big((v \mp y_0)/(u - x_0)\big)^2 - (u + x_0)$ and use $v^2 + y_0^2 = u^3 + x_0^3 + a(u + x_0) + 2b$ together with $(u + x_0)(u - x_0)^2 = u^3 - u^2 x_0 - u x_0^2 + x_0^3$; (2) is a similar computation (it is part of the formalized algebraic core, Section 5).

3 Main results

Theorem 1 (B, k^2-interpolation)**.** *Let W be a short Weierstrass curve over a field F with $2 \neq 0 \neq 3$ in F, and P a point of odd order r. If $g \in F[X]$ satisfies $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, then $r \le 6 \deg g + 5$; that is, $\deg g \ge (r-5)/6$.*

For an elliptic curve over $\mathbb{F}_p$ and P of prime order $r > 2$ this is the statement recorded as HYP-eda6732a (the ledger statement assumes r prime; the blind review REV-83ec2fc9 notes that "r prime" can be weakened to "r odd", which is the formalized form).

Theorem 2 (C, linear interpolation)**.** *Let W be short Weierstrass over F with $2 \neq 0$, P a point with $\mathrm{ord}(P) = r$, and $2N < r$. If $g \in F[X]$ satisfies $g(x(kP)) = k$ in F for $1 \le k \le N$, then $N \le 3 \deg g + 2$. In particular, for $N = (r-1)/2$ (r odd), $r \le 6 \deg g + 5$.*

(HYP-42f91290.) The assumption $3 \neq 0$ is not needed.

Theorem 3 (D, partial agreement)**.** *Let W be short Weierstrass over F with $2 \neq 0$, $\mathrm{ord}(P) = r$, $2N < r$ (equivalently $N \le (r-1)/2$ for r odd), $A \subseteq \{1, \dots, N\}$, and $g \in F[X]$ with $g(x(kP)) = k^2$ for all $k \in A$. For $j \ge 1$ let $S_j = \#\{k \in A : k - j \in A,\ k + j \in A\}$ (the k with $j < k$ automatically). If $\deg g \ge 1$ or $j \neq 0$ in F, then $S_j \le 3 \deg g$.*

(HYP-a1dd5501, part (2); the main inequality only.) The side condition is needed over general fields: if g is constant and $j = 0$ in F then $g(x(kP)) = k^2$ on A with $k \pm j \in A$ can hold with $\deg g = 0$ and $S_j > 0$. Over $\mathbb{F}_p$ the condition excludes only constant g with $p \mid j$, and then $S_j = 0$ for the following reason (*not* formalized in Lean): $S_j \ge 1$ requires $k - j \ge 1$ and $k + j \le N$, so $N \ge 2j + 1 \ge 2p + 1$, whence $r > 2N \ge 4p + 2$, contradicting the Hasse bound $r \le \#E(\mathbb{F}_p) \le p + 1 + 2\sqrt{p}$. Hence over $\mathbb{F}_p$ the unconditional informal statement $S_j \le 3 \deg g$ holds. (One must not argue $j \le N < p$; that is false in general, e.g. $y^2 = x^3 + 2x + 1$ over $\mathbb{F}_3$ with $r = 7$, $N = 3$.)

Remark 4 (Hypotheses and the refuted transcription). The hypothesis $2N < r$ in Theorem 3 cannot be dropped. The ledger record HYP-5da6025e (status: refuted) contains a transcription of part (2) with $N < \min(p, r)$ in place of $N \le (r-1)/2$; the blind review REV-ac47cc67 produced the counterexample $y^2 = x^3 + 5$ over $\mathbb{F}_7$, $P = (3, 2)$ of order $7 = p$, $A = \{1, \dots, 6\}$, $g = 5X$, which satisfies $g(x(kP)) = k^2$ for $k = 1, \dots, 6$ with $\deg g = 1 < S_1/3 = 4/3$. The hypothesis $N \le (r-1)/2$ had been present in the original agent report and was lost in a transcription error by the principal investigator; the corrected statement is HYP-a1dd5501 and is what is formalized here. Ledger parts of HYP-a1dd5501 not covered here: part (1) for rational functions on E (reviewed, not formalized), and the density corollary $\deg g \ge ((3\delta - 2)N - 2)/3$ for $\delta > 2/3$ (reviewed, not formalized).

Remark 5 (Background: Theorem A). The ledger also records (HYP-adb581ff, status *open*, reviewed, not formalized) a result for rational functions f on E with $f(kP) = k$ for $1 \le k \le N \le r-1$: the pole-divisor degree is at least $(N-1)/2$, and polynomial total degree at least $(N-1)/6$. It is recorded as known in kind (Section 6) and is cited only as background.

4 Proofs

Assume $m = \deg g \ge 1$ for the moment; constant g is treated at the end. Fix an affine point $P_0 = (x_0, y_0)$ ($P_0 = P$ in Theorems B, C; $P_0 = jP$ in Theorem D) and put $d = u - x_0$.

The cleared second-difference polynomial. For a variable u let $s_\pm$ be the (formal) roots of $z^2 - e_1 z + e_2$ where $e_1 = (s_+ + s_-)$ and $e_2 = s_+ s_-$ are the rational functions in u given by (1) and (2): $e_1 = n_1/d^2$, $e_2 = n_2/d^2$ with n_1, n_2 polynomials of degree ≤ 2. Write $g = \sum_{i \le m} c_i X^i$. Then $g(s_+) + g(s_-) = \sum_i c_i p_i$ with $p_i = s_+^i + s_-^i$ a polynomial with integer coefficients in e_1, e_2 whose monomials $e_1^\alpha e_2^\beta$ satisfy $\alpha + 2\beta = i$ (Newton–Girard). Multiplying a monomial by d^{2m} gives $n_1^\alpha n_2^\beta d^{2(m-\alpha-\beta)}$, a polynomial of degree $\le 2(\alpha+\beta) + 2(m-\alpha-\beta) = 2m$, because $\alpha + \beta \le i \le m$. Define

$$G_c(u) = d^{2m}\big[g(s_+) + g(s_-)\big] - 2d^{2m} g(u) - c\, d^{2m} \in F[u],$$

for a constant c. The first term has degree $\le 2m$, the third degree $2m$, and $-2d^{2m}g(u)$ has degree exactly $3m$ with leading coefficient $-2\,\mathrm{lc}(g) \ne 0$ (here $2 \ne 0$ is used). Since $2m < 3m$ for $m \ge 1$:

$$\deg G_c = 3m \quad \text{for every constant } c, \quad \text{in particular } G_c \ne 0. \tag{3}$$

Roots. Suppose $Q = kP \ne P_0$ in x-coordinate, $x(Q) \ne x_0$, with $Q \pm P_0$ affine. Then evaluating at $u = x(Q)$, the quantities $s_\pm$ are $x(Q \pm P_0)$ by the identities of Section 2, so $G_c(x(Q)) = d^{2m}\left[g(x(Q+P_0)) + g(x(Q-P_0)) - 2g(x(Q)) - c\right]$ with $d \ne 0$.

Theorem B. Take $P_0 = P$, $c = 2$. For $2 \le k \le (r-3)/2$ we have $1 \le k-1$ and $k+1 \le (r-1)/2$, so all three values of g are prescribed and the bracket equals $(k+1)^2 + (k-1)^2 - 2k^2 - 2 = 0$. The values $x(kP)$, $2 \le k \le (r-3)/2$, are pairwise distinct (injectivity) and differ from $x(P)$ (since $k \ge 2$), so G_2 has at least $(r-5)/2$ distinct roots. By (3), $(r-5)/2 \le 3m$, i.e. $r \le 6m+5$. Constant $g = c_0$: then $g(x(P)) = 1$ and $g(x(2P)) = 4$ force $3 = 0$ if $(r-1)/2 \ge 2$, excluded by $3 \ne 0$; and $r \le 5$ is trivial. Hence the bound holds.

Theorem C. Take $P_0 = P$, $c = 0$. For $2 \le k \le N-1$ the bracket is $(k+1) + (k-1) - 2k = 0$, giving $N-2$ distinct roots (injectivity as $2N < r$), so $N - 2 \le 3m$. If g is constant and $N \ge 2$, then $g(x(P)) = 1 \ne 2 = g(x(2P))$, contradiction; so $N \le 1 \le 3m + 2$.

Theorem D. Take $P_0 = jP$ (note $j < k \le N < r$). For $k \in C_j = \{k \in A : j < k,\ k \pm j \in A\}$ the bracket with $c = 2j^2$ is $(k+j)^2 + (k-j)^2 - 2k^2 - 2j^2 = 0$. The values $x(kP)$ are distinct for distinct $k \in C_j$ and differ from $x(jP)$ because $k \ne j$ and $k + j \le 2N < r$, so $|C_j| \le 3m$ by (3). If $g = c_0$ is constant, the bracket is $-2j^2 \ne 0$ whenever $j \ne 0$ in F, so G has no roots and $C_j = \emptyset$. □

Exact degree. The computation above shows that, for the second-difference identity, the cleared polynomial has degree exactly $3 \deg g$ and leading coefficient $-2\,\mathrm{lc}(g)$; for Theorem C this is recorded in HYP-42f91290 and agrees with the numerical checks of the ledger.

5 Formal verification

Environment. Toolchain `leanprover/lean4:v4.35.0` (file `lean-toolchain`); Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` (`lakefile.toml`, with

`-DwarningAsError=true`). Source commits recorded in the proof records: Theorem B, PRF-34538f75, commit `cc8dc791a0bd74f89b6cd614898a5773abdc500a` (`lake build`: 3163 jobs, 0 warnings); Theorem C, PRF-35d1387a, commit `8191c0f69398b632fc2c7c7efbfdcdb9e2634b7b` (3164 jobs, 0 warnings); Theorem D, PRF-5357b698, commit `84ddbad82b2f1412e851cf1ab6012b6650bba544` (3165 jobs, 0 warnings). Files: `MathResearch/ECDLP/Interpolation.lean` (generic polynomial core), `CurveX.lean`, `LinearInterp.lean`, `PartialAgreement.lean`. All three proofs use Mathlib's `WeierstrassCurve.Affine.Point`; the curve-specific input is the x-addition identities (`CurveX.sum_prod_identity`, `CurveX.smul_relations`) and injectivity of $k \mapsto x(kP)$.

Statements (verbatim from the repository). Here `xCoord` is the x-coordinate function on `W.Point` and `F` a field.

```
theorem degree_bound_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (h3' : (3 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (hodd : r % 2 = 1) (g : F[X])
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ (r - 1) / 2 → g.eval (xCoord (k • P)) = (k : F) ^ 2)
     :
    r ≤ 6 * g.natDegree + 5
```

```
theorem degree_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
     = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (N : ℕ) (hN : 2 * N < r) (g : Polynomial F)
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ N → g.eval (xCoord (k • P)) = (k : F)) :
    N ≤ 3 * g.natDegree + 2
```

```
theorem partial_agreement_degree_bound {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (A : Finset ℕ) (hA : ∀ k ∈ A, 1 ≤ k ∧ k ≤ N) (g : F[X])
    (hg : ∀ k ∈ A, g.eval (xCoord (k • P)) = (k : F) ^ 2) (j : ℕ) (hj : 1 ≤ j)
    (hjg : 1 ≤ g.natDegree ∨ (j : F) ≠ 0) :
    (A.filter (fun k => j < k ∧ k - j ∈ A ∧ k + j ∈ A)).card ≤ 3 * g.natDegree
```

The full names are `MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass`, `MathResearch.ECDLP.CurveX.degree_bound_linear_short_weierstrass` and `MathResearch.ECDLP.CurveX.partial_agreement_degree_bound`.

Correspondence to the informal statements. (i) The formal statements are for an arbitrary field F, not only $\mathbb{F}_p$, and for any point P of the given additive order; r need not be prime. Theorem B needs r odd, $2 \neq 0 \neq 3$; Theorems C and D need neither $3 \neq 0$ nor r odd (the condition $2N < r$ is equivalent to $N \leq (r-1)/2$ for r odd). (ii) Since `W : Affine F` is not assumed nonsingular, the review REV-1f44f52b notes that the results also cover nonsingular points of singular cubics. (iii) Theorem D carries the extra hypothesis `hjg`; the Hasse-bound argument that removes it over $\mathbb{F}_p$ (Section 3) is *not* formalized. (iv) The formal statements are degree bounds only; they assert no existence of g and no complexity. REV-1f44f52b and REV-6c9bfbc1 accepted the correspondence of B and C; the independent review REV-685408b5 (formal correctness only; no complexity or novelty verification) verified the correspondence for D.

Trust base. `#print axioms` on all three declarations (and on `smul_relations`, `sum_prod_identity`) reports exactly `[propext, Classical.choice, Quot.sound]` (evidence: `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-b585f279-fd16-4cca-b3a1-c2484d545078.evidence`). Trust therefore rests on the Lean

kernel, Mathlib at the pinned revision, and the faithfulness of the statements to the informal ones (reviewed by AI agents of the same provider; independence is partial).

Reproduction. At a checkout of the repository: `lake build` (pinned toolchain and Mathlib), followed by `#print axioms MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass` and likewise for the other two names (see `research/ecdlp/proofs/checks/Round8Axioms.lean` for the pattern).

Empirical sanity checks (not part of the proof). The ledger records toy-size numerical checks: the x-addition identity 2519 times with no failure (Theorem B record); 208 toy cases without violation (REV-83ec2fc9); for Theorem C 533 curves with $p \leq 211$, then 12,459 cases including extension fields (REV-6c9bfbc1); 1242 toy points for B (REV-1f44f52b); 739,847 pairs (A, j) for D (a weak check). These tests do not constitute proofs and are not needed given the Lean proofs; they indicate the bounds are far from tight in tests.

6 Relation to prior work and novelty status

Lange and Winterhof (COCOON 2002, LNCS 2387, 137–143 [1]; AAECC-15 2003, LNCS 2643, 51–60 [2]) and Satoh (WCC 2005, LNCS 3969 [3]) prove degree lower bounds for polynomial interpolation of the elliptic curve discrete logarithm and related maps. **The exact statements of these results were not accessible to the laboratory**: only bibliographic records and abstracts were seen, and full texts were not accessed. We therefore cannot compare constants or hypotheses. Our bounds may be known or weaker. Theorem A is recorded in the ledger as known in kind; the laboratory's novelty checks (ECDLP-22) found no result of the form of Theorems B–D in the accessible material, but this is absence of evidence, not a novelty claim. The review of Theorem C judges it probably not novel. Novelty of all three is *unsettled*.

Claimed contribution: a machine-checked formalization (Lean 4, Mathlib, standard axioms only) of the three degree bounds, with explicit, reviewed hypotheses (including the corrected hypothesis $2N < r$ of Theorem D and its generality over arbitrary fields). Bibliographic entries above carry only the data present in the ledger (author surnames, venue, volume, pages); the publication year of [3] is not recorded in the ledger and is entered as unverified.

7 Limitations

- Degree is not time. A lower bound on $\deg g$ for polynomials in one variable $x(kP)$ does not bound the running time of any algorithm, and we draw no bit-complexity consequence.
- The results concern interpolation from x-coordinates only, with exact agreement on a range (B, C) or a subset (D); they say nothing about approximate or probabilistic computation or about other coordinates.
- No existence statement: we do not claim that interpolating polynomials exist, nor give a matching upper bound or tightness result.
- The ledger's rational-function versions (Theorem A, part (1) of the partial-agreement hypothesis) and the density corollary are not formalized.
- Theorem D's extra hypothesis is removed over $\mathbb{F}_p$ only by an informal Hasse-bound argument.
- The universal $O(n \log n)$ ECDLP target remains open; nothing here improves on any known ECDLP complexity.

- Novelty is unsettled (Section 6); no human has verified the mathematics, the Lean-to-statement correspondence, or the bibliography.

Provenance note. This report was produced by an AI research laboratory (this draft by a worker agent reporting model identity Sonnet 5.5, `claude-sonnet-5-5`, not verifiable from runtime metadata) and reviewed by AI agents; no human verification has yet taken place. It is an internal technical report and not a submission.

References

[1] Lange and Winterhof. Polynomial interpolation of the elliptic curve and XTR discrete logarithm. In *COCOON 2002*, volume 2387 of *Lecture Notes in Computer Science*, pages 137–143. Springer, 2002. Bibliographic data from ledger records; first names, editors and DOI not recorded; full text not accessed.

[2] Lange and Winterhof. Interpolation of the elliptic curve Diffie–Hellman mapping. In *AAECC-15*, volume 2643 of *Lecture Notes in Computer Science*, pages 51–60. Springer, 2003. Bibliographic record only; theorem statements not seen.

[3] Satoh. On degrees of polynomial interpolations related to elliptic curve cryptography. In *WCC 2005*, volume 3969 of *Lecture Notes in Computer Science*, pages 155–163. Springer, 2006. Abstract only (paywalled); year of the volume and first name not recorded in the ledger, year entered as the volume's publication year (unverified).